



## Title

**kdrobust** — Kernel Density Estimation with Robust Bias-Corrected Confidence Intervals and Inference Procedures.

## Syntax

```
kdrobust varname [if] [in] [, eval(gridvar) neval(#) h(#) b(#) rho(#)
kernel(kernelfn) bwselect(bwmethod) bwcheck(#) imsegrid(#) level(#)
separator(#) genvars plot graph_options(gphopts) ]
```

## Description

**kdrobust** implements kernel density point estimators with robust bias-corrected confidence intervals and inference procedures developed in [Calonico, Cattaneo and Farrell \(2018\)](#). See also [Calonico, Cattaneo and Farrell \(2022\)](#) for related optimality results. It also implements other estimation and inference procedures available in the literature.

A detailed introduction to this command is given in [Calonico, Cattaneo and Farrell \(2019\)](#).

Companion command is: [kdbwselect](#) for data-driven bandwidth selection.

Related software useful for empirical analysis is described in the following website:

<https://nppackages.github.io/>

Requires Stata 14 or later.

## Options

**eval**(gridvar) specifies the grid of evaluation points for *xvar*. By default it uses 30 quantile-spaced points (deciles 0.1 through 0.9 in equal steps) over the support of *xvar*.

**neval**(#) specifies the number of quantile-spaced evaluation points used to estimate the density. Default is 30 evaluation points.

**h**(hvar) specifies the main bandwidth (*h*) used to construct the point estimator for each evaluation point. If not specified, it is computed by the companion command [kdbwselect](#).

**b**(bvar) specifies the bias bandwidth (*b*) used to construct the bias-correction estimator for each evaluation point. If not specified, it is computed by the companion command [kdbwselect](#).

**rho**(#) specifies the value of *rho*, so that the bias bandwidth *b* equals  $b=h/\rho$ . Default is **rho(1)** if *h* is specified but *b* is not.

**kernel**(kernelfn) specifies the kernel function used to construct the kernel density estimator(s). Options are: **epanechnikov**, and **uniform**. Default is **kernel(epanechnikov)**.

**bwselect**(bwmethod) bandwidth selection procedure to be used. By default it computes both *h* and *b*, unless *rho* is specified, in which case it only computes *h* and sets  $b=h/\rho$ . Options are:

**mse-dpi** second-generation DPI implementation of MSE-optimal bandwidth. Default choice.

**imse-dpi** second-generation DPI implementation of IMSE-optimal bandwidth.

**imse-rot** ROT implementation of IMSE-optimal bandwidth.

**ce-dpi** second generation DPI implementation of CE-optimal bandwidth.

**ce-rot** ROT implementation of CE-optimal bandwidth.

Note: MSE = Mean Square Error; IMSE = Integrated Mean Squared Error; CE = Coverage Error; DPI = Direct Plug-in; ROT = Rule-of-Thumb.

Default is **bwselect(mse-dpi)**. For details on implementation see [Calonico, Cattaneo and Farrell \(2019\)](#).

**bwcheck**(#) specifies an optional positive integer so that the selected bandwidth is enlarged to have at least # effective observations available for each evaluation point.

**imsegrid**(#) number of evaluations points used to compute the IMSE bandwidth selector. Default is 30 points.

**level**(#) specifies confidence level for confidence intervals. Default is **level(95)**.

**separator**(#) draws separator line after every # variables; default is **separator(5)**.

**plot** generates the kernel density plot.

**genvars** generates new variables storing the following results.

- kdrobust\_eval** evaluation points.
- kdrobust\_h** bandwidth h.
- kdrobust\_b** bandwidth b.
- kdrobust\_N** effective sample size.
- kdrobust\_tau\_us** conventional kernel density estimate.
- kdrobust\_se\_us** conventional standard error.
- kdrobust\_tau\_bc** bias-corrected kernel density estimate.
- kdrobust\_se\_rb** robust standard error for the kernel density estimator.
- kdrobust\_ci\_l\_rb** lower end value of the robust confidence interval.
- kdrobust\_ci\_r\_rb** upper end value of the robust confidence interval.

**graph\_options**(*gphopts*) specifies graphical options to be passed on to the underlying graph command.

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#### **Example: Cholesterol Trial Data**

```

Setup
    . use nprobust_data.dta

Kernel density estimates for baseline cholesterol in the control group
    . kdrobust chol1 if t==0

Same as above, but generating a plot and the corresponding output variables
    . kdrobust chol1 if t==0, plot genvars

```

#### **Stored results**

**kdrobust** stores the following in **e()**:

##### Scalars

|                 |                                     |
|-----------------|-------------------------------------|
| <b>e(N)</b>     | sample size after listwise deletion |
| <b>e(level)</b> | confidence level                    |

##### Macros

|                    |                              |
|--------------------|------------------------------|
| <b>e(cmd)</b>      | command name                 |
| <b>e(xvar)</b>     | name of independent variable |
| <b>e(bwselect)</b> | bandwidth selection choice   |
| <b>e(kernel)</b>   | kernel choice                |

##### Matrices

|                  |                   |
|------------------|-------------------|
| <b>e(Result)</b> | estimation result |
|------------------|-------------------|

#### **References**

- Calonico, S., M. D. Cattaneo, and M. H. Farrell. 2018. [On the Effect of Bias Estimation on Coverage Accuracy in Nonparametric Inference](#). *Journal of the American Statistical Association*, 113(522): 767-779.
- Calonico, S., M. D. Cattaneo, and M. H. Farrell. 2019. [nprobust: Nonparametric Kernel-Based Estimation and Robust Bias-Corrected Inference](#). *Journal of Statistical Software*, 91(8): 1-33. doi: 10.18637/jss.v091.i08.

Calonico, S., M. D. Cattaneo, and M. H. Farrell. 2022. Coverage Error Optimal Confidence Intervals for Local Polynomial Regression. *Bernoulli*, 28(4): 2998-3022.

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